

# XING SHENG

## Professor

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### Education

- *Ph.D., Materials Science and Engineering, Massachusetts Institute of Technology, 2012*  
Advisor: Lionel C. Kimerling
- *B. Eng., Materials Science and Engineering, Tsinghua University, 2007*

### Professional Experiences

- *Professor, Department of Electronic Engineering, Tsinghua University, present*  
also affiliated with: IDG/McGovern Institute for Brain Research at Tsinghua University
- *Postdoctoral Associate, University of Illinois at Urbana-Champaign, 2012–2015*  
Advisor: John A. Rogers

### Research Interests

- Non-conventional Optoelectronics for Biomedical Applications
- Optical Neural Interfaces
- Biocompatible and Biodegradable Photonics

### Teaching Experience

- Course Instructor at Tsinghua
  - 20230313 “Foundation of Solid State Physics”
  - 80230992 “Principles of Micro- and Nanofabrication for Electronic and Photonic Devices”
  - 80231001 “Laboratory of Micro- and Nanofabrication for Electronic and Photonic Devices”
  - 60230072 “Academic Writings and Presentations for Electrical Engineering”
- Worked as a guest lecturer and a teaching assistant for multiple courses at Tsinghua, MIT and UIUC
- Supervised > 30 undergraduate and > 10 graduate students at Tsinghua, MIT and UIUC

Lecture notes for these courses have been posted online and received widespread attention:

<https://shengxingstars.github.io/www/teaching.html>

## Publications

### Peer-Reviewed Journals:

Google Scholar: <https://scholar.google.com/citations?hl=en&user=bS9skH4AAAAJ>

#co-first author, \*corresponding author

1. Y. Xie, Y. Peng, J. Guo, M. Liu, B. Zhang, L. Yin, H. Ding\*, **X. Sheng\***, “Materials and Devices for High-Density, High-Throughput Micro-Electrocorticography Arrays”, *Fundamental Research* **5**, 17–28 (2025) (**Front Cover**).
2. **X. Sheng\***, Y. Sui, L. Li, J. A. Rogers, “Multimodal technologies for neural modulation and sensing”, *Fundamental Research* **5**, 1–2 (2025).
3. L. Wang, T. Zhang, J. Lei, S. Wang, Y. Guan, K. Chen, C. Li, Y. Song, W. Li, S. Wang, Z. Jia, S. Chen, J. Bai, B. Yu, C. Yang, P. Sun, Q. Wang, **X. Sheng**, J. Peng, Y. Fan\*, L. Wang\*, M. Zhang\*, Y. Wang\*, L. Yin\*, “A biodegradable and restorative peripheral neural interface for the interrogation of neuropathic injuries”, *Nature Communications* **16**, 1716 (2025).
4. H. Wen, L. Kong, X. Zhu, Y. Miao, **X. Sheng**, X. Chen, Y. Liu\*, P. Chen\*, “Mutually reinforcing and transpiration-dependent propagation of H<sub>2</sub>O<sub>2</sub> and variation potential in plants revealed by fiber organic electrochemical transistors”, *The Innovation* **6**, 100800 (2025).
5. J. Tian, J. Li, Z. Wu, L. Yin, **X. Sheng**, Z. Li, H. Wang\*, H. Wang\*, Y. Zhao\*, “Photo-driven electronic skin enable activation of calcium channel for refractory wound healing”, *Nano Today* **62**, 102697 (2025).
6. X. Cai#, H. Zhang#, P. Wei#, Q. Liu#, D. Sheng, Z. Li, B. Zhang, G. Tang, W. Zhao, Z. Ye, Z. Xue, Y. Xie, Y. Dai, C. Wang, Y. Wang, X. Fu, L. Yin, H. Peng, H. Ding\*, G. Zhao\*, **X. Sheng\***, “A wireless optoelectronic probe to monitor oxygenation in deep brain tissue”, *Nature Photonics* **18**, 492–500 (2024) (**Front Cover**).
7. X. Fu#, Z. Hu#, W. Li, L. Ma, J. Chen, M. Liu, J. Liu, S. Hu, H. Wang, Y. Huang, G. Tang, B. Zhang, X. Cai, Y. Wang, L. Li, J. Ma, S.-H. Shi, L. Yin, H. Zhang, X. Li\*, **X. Sheng\***, “A Silicon Diode-based Optoelectronic Interface for Bidirectional Neural Modulation”, *Proceedings of the National Academy of Sciences USA* **121**, e2404164121 (2024).
8. J. Chen, H. Ding\*, Y. Li, **X. Sheng\***, “Numerical Design of a Micro-LED based Optogenetic Stimulator for Visual Cortical Prosthesis”, *IEEE Journal of Selected Topics in Quantum Electronics* **30**, 7200308 (2024) (**Front Cover**).
9. J. Chen, H. Ding\*, **X. Sheng\***, “Advanced Manufacturing of Microscale Light-Emitting Diodes and Their Use in Displays and Biomedicine”, *Journal of Information Display* **25**, 1–12 (2024) (**Invited**).
10. Q. Du, H. Dai, Y. Sui, **X. Sheng\***, “Recent Progress on Multimodal Technologies for Neural Modulation and Sensing”, *Bulletin of National Natural Science Foundation of China* **38**, 296–302 (2024) (**Front Cover**).  
杜全生, 戴慧, 眭亚楠, **盛兴\***, 多模态神经调控与检测技术研究进展, 中国科学基金, **38**, 296–302 (2024).
11. L. Li, B. Zhang, W. Zhao, D. Sheng, L. Yin, **X. Sheng\***, D. Yao\*, “Multimodal Technologies for Closed-Loop Neural Modulation and Sensing”, *Advanced Healthcare Materials* **13**, 2303289 (2024) (**Invited**).
12. Z. Xue, W. Chou, Y. Xu, Z. Cheng, X. Ren, T. Sun, W. Tong, Y. Xie, J. Chen, N. Zhang, **X. Sheng**, Y. Wang, H. Zhao, J. Yang\*, H. Ding\*, “Battery-free Optoelectronic Patch for Photodynamic and Light Therapies in Treating Bacteria-infected Wounds”, *Biosensors and Bioelectronics* **261**, 116467 (2024).

13. B. Yu, J. Bai, Y. Guan, X. Huang, L. Liang, Z. Ren, X. Song, T. Zhang, C. Yang, F. Dai, X. Wang, **X. Sheng**, J. Peng, L. Wang\*, Y. Wang\*, L. Yin\*, “Fully biodegradable and self-powered nerve guidance conduit based on zinc-molybdenum batteries for peripheral nerve repair”, *Biosensors and Bioelectronics* **263**, 116578 (2024).
14. L. Wang, S. Liu, W. Zhao, J. Li, H. Zeng, S. Kang, **X. Sheng**, L. Wang\*, Y. Fan\*, L. Yin\*, “Recent Advances in Implantable Neural Interfaces for Multimodal Electrical Neuromodulation”, *Advanced Healthcare Materials* **13**, 2303316 (2024) (*Invited*).
15. C. Hu, L. Wang, S. Liu, **X. Sheng**, L. Yin\*, “Recent Development of Implantable Chemical Sensors Utilizing Flexible and Biodegradable Materials for Biomedical Applications”, *ACS Nano* **18**, 3969–3995 (2024).
16. P. Sun, C. Li, C. Yang, M. Sun, H. Hou, Y. Guan, J. Chen, S. Liu, K. Chen, Y. Ma, Y. Huang, X. Li, H. Wang, L. Wang, S. Chen, H. Cheng, W. Xiong, **X. Sheng**, M. Zhang, J. Peng, S. Wang\*, Y. Wang\*, L. Yin\*, “A Biodegradable and Flexible Neural Interface for Transdermal Optoelectronic Modulation and Regeneration of Peripheral Nerves”, *Nature Communications* **15**, 4721 (2024).
17. X. Cheng, W. Li, Y. Wang, K. Weng\*, Y. Xing, Y. Huang, **X. Sheng**, J. Yao\*, H. Zhang\*, J. Li\*, “Highly Branched Au Superparticles as Efficient Photothermal Transducers for Optical Neuromodulation”, *ACS Nano* **18**, 29572–29584 (2024).
18. X. Wang, A. Aziz, **X. Sheng**, L. Wang\*, L. Yin\*, “Bioresorbable neural interfaces for bioelectronic medicine”, *Current Opinion in Biomedical Engineering* **32**, 100565 (2024).
19. J. Liu, Z. Li, M. Sun, L. Zhou, X. Wu, Y. Lu, Y. Shao, C. Liu, N. Huang, B. Hu, Z. Wu, C. You, L. Li, M. Wang, L. Tao, Z. Di, **X. Sheng**, Y. Mei\*, E. Song\*, “Flexible Bioelectronic Systems with Large-Scale Temperature Sensor Array for Monitoring and Treatments of Localized Wound Inflammation”, *Proceedings of the National Academy of Sciences USA* **121**, e2412423121 (2024).
20. Y. M. Song, Y. Chai, D.-H. Kim, **X. Sheng**, D. Son, S. Kim, “Editorial Interview Flexible Optoelectronics”, *IEEE Journal of Selected Topics in Quantum Electronics* **30**, 1000403 (2024).
21. Y. Huang#, Y. Cui#, H. Deng#, J. Wang, R. Hong, S. Hu, H. Hou, Y. Dong, H. Wang, J. Chen, L. Li, Y. Xie, P. Sun, X. Fu, L. Yin, W. Xiong, S.-H. Shi, M. Luo, S. Wang\*, X. Li\*, **X. Sheng\***, “Bioresorbable Thin-Film Silicon Diodes for the Optoelectronic Excitation and Inhibition of Neural Activities”, *Nature Biomedical Engineering* **7**, 486–498 (2023) (*Front Cover*).
22. H. Wang, J. Tian, Y. Jiang, S. Liu, J. Zheng, N. Li, G. Wang, F. Dong, J. Chen, Y. Xie, Y. Huang, X. Cai, X. Wang, W. Xiong, H. Qi, L. Yin, Y. Wang\*, **X. Sheng\***, “A 3D Biomimetic Optoelectronic Scaffold Repairs Cranial Defects”, *Science Advances* **9**, abq7750 (2023).
23. H. Ding\*, Y. Peng, G. Lv, Y. Xie, J. Chen, Z. Shi, Y. Deng, L. Yin, J. Yang, Y. Wang, **X. Sheng\***, “Heterogeneous Integration of Thin-Film Organic and Inorganic Devices for Optical based Bioelectrical and Chemical Sensing”, *IEEE Journal of Selected Topics in Quantum Electronics* **29**, 5200107 (2023) (*Invited*).
24. **X. Sheng\***, W. Zhao, L. Li, Y. Huang, H. Ding, “Foundation of Brain-Machine Interfaces: Neurons and Diodes”, *Chinese Journal of Lasers* **50**, 0907301 (2023) (*Invited*) (*Front Cover*).  
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25. Z. Shi#, X. Gao#, X. Cai, H. Zhao, X. Wang, L. Zhao, L. Yin, H. Ding\*, **X. Sheng\***, “Fully Implantable and Retrievable Upconversion Waveguides for Photodynamic Therapy in Deep Tissue”, *Advanced Optical Materials* **11**, 2300689 (2023) (*Inside Front Cover*).
26. K. Zhang, W. Zhao, **X. Sheng\***, “Ion-Gated Tungsten Oxide Based Electrochemical Transistors with Subthreshold Slopes Approaching the Thermodynamic Limit”, *Applied Physics A* **129**, 728 (2023).
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29. S. Liu, Q. Yu, R. Guo, K. Chen, J. Xia, Z. Guo, L. He, Q. Wu, L. Liu, Y. Li, B. Zhang, L. Lu, **X. Sheng**, J. Zhu, L. Zhao, H. Qi, K. Liu\*, L. Yin\*, “A Biodegradable, Adhesive and Stretchable Hydrogel and Potential Applications for Allergic Rhinitis and Epistaxis”, *Advanced Healthcare Materials* **12**, 2302059 (2023).
30. L. Kong, H. Wen, Y. Luo, X. Chen, **X. Sheng**, Y. Liu\*, P. Chen\*, “Dual-Conductive and Stiffness-Morphing Microneedle Patch Enables Continuous in Planta Monitoring of Electrophysiological Signal and Ion Fluctuation”, *ACS Applied Materials & Interfaces* **15**, 43515–43523 (2023).
31. X. Huang, H. Hou, B. Yu, J. Bai, Y. Guan, L. Wang, K. Chen, X. Wang, P. Sun, Y. Deng, S. Liu, X. Cai, Y. Wang, J. Peng, **X. Sheng**, W. Xiong\*, L. Yin\*, “Fully Biodegradable and Long-Term Operational Primary Zinc Batteries as Power Sources for Electronic Medicine”, *ACS Nano* **17**, 5727–5739 (2023).
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33. F. Dai, Q. Geng, T. Hua, **X. Sheng**, L. Yin\*, “Organic Biodegradable Piezoelectric Materials and Their Potential Applications as Bioelectronics”, *Soft Science* **3**, 7 (2023) (**Invited**).
34. S. Liu, X. Wang, S. Liu, Y. Deng, B. Zhao, H. Wang, **X. Sheng**, L. Zhao, L. Wang\*, P. Zhang\*, L. Yin\*, “Laser-Triggered Degradation of Silicon Circuits by Lithiation and Moisture Uptake for On-Demand Transient Electronics”, *Advanced Engineering Materials* **25**, 2300213 (2023).
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36. Y. Luo, X. Chen\*, *et al.*, “Technology Roadmap for Flexible Sensors”, *ACS Nano* **17**, 5211–5295 (2023).
37. L. Li#, L. Lu#, Y. Ren#, G. Tang#, Y. Zhao, X. Cai, Z. Shi, H. Ding, C. Liu, D. Cheng, Y. Xie, H. Wang, X. Fu, L. Yin, M. Luo\*, **X. Sheng\***, “Colocalized, Bidirectional Optogenetic Modulations in Freely Behaving Mice with a Wireless Dual-Color Optoelectronic Probe”, *Nature Communications* **13**, 839 (2022).
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39. H. Ding\*, G. Lv, X. Cai, J. Chen, Z. Cheng, Y. Peng, G. Tang, Z. Shi, Y. Xie, X. Fu, L. Yin, J. Yang, Y. Wang, **X. Sheng\***, “An Optoelectronic Thermometer based on Microscale Infrared-to-Visible Conversion Devices”, *Light: Science & Applications* **11**, 130 (2022).
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41. R. Nazempour#, B. Zhang#, Z. Ye, L. Yin, X. Lv, **X. Sheng\***, “Emerging Applications of Optical Fiber-Based Devices for Brain Research”, *Advanced Fiber Materials* **4**, 24–42 (2022) (**Invited Review**).
42. D. Kong, Y. Zhang, D. Cheng, E. Wang, K. Zhang, H. Wang, K. Liu, L. Yin\*, **X. Sheng\***, “Heteroepitaxy of Large-Area, Monocrystalline Lead Halide Perovskite Films on Gallium Arsenide”,

*ACS Applied Materials & Interfaces* **14**, 52508–52515 (2022).

43. D. Kong#, K. Zhang#, J. Tian, L. Yin\*, **X. Sheng\***, “Biocompatible and Biodegradable Light-Emitting Materials and Devices”, *Advanced Materials Technologies* **7**, 2100006 (2022) (**Invited Review**).
44. Y. Deng, H. Qi, Y. Ma, S. Liu, M. Zhao, Z. Guo, Y. Jie, R. Zheng, J. Jing, K. Chen, H. Ding, G. Lv, K. Zhang, R. Li, H. Cheng, L. Zhao, **X. Sheng**, M. Zhang\*, L. Yin\*, “A flexible and highly sensitive organic electrochemical transistor-based biosensor for continuous and wireless nitric oxide detection”, *Proceedings of the National Academy of Sciences USA* **119**, e2208060119 (2022).
45. S. Lu, Z. Fu, F. Li, K. Weng, L. Zhou, L. Zhang, Y. Yang, H. Qiu, D. Liu, W. Qing, H. Ding, **X. Sheng**, M. Chen, X. Tang, L. Duan, W. Liu, L. Wu, Y. Yang, H. Zhang\*, J. Li, “Beyond a Linker: The Role of Photochemistry of Crosslinkers in the Direct Optical Patterning of Colloidal Nanocrystals”, *Angewandte Chemie* **134**, e202202633 (2022).
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47. H. Ding, G. Lv, Z. Shi, D. Cheng, Y. Xie, Y. Huang, L. Yin, J. Yang, Y. Wang, **X. Sheng\***, “Optoelectronic Sensing of Biophysical and Biochemical Signals based on Photon Recycling of a micro-LED”, *Nano Research* **14**, 3208–3213 (2021) (**Invited**) (**Front Cover**).
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51. R. Nazempour, Q. Zhang, C. Liu, **X. Sheng\***, “Design of Silicon Photonic Structures for Multi-site, Multi-Spectral Optogenetics in the Deep Brain”, *IEEE Photonics Journal* **12**, 4200107 (2020).
52. D. Kong#, D. Cheng#, X. Wang, K. Zhang, H. Wang, K. Liu, H. Li, **X. Sheng\***, L. Yin\*, “Solution Processed Lead-free Cesium Titanium Halide Perovskites and Their Structural, Thermal and Optical Characteristics”, *Journal of Materials Chemistry C* **8**, 1591–1597 (2020) (**Inside Front Cover**).
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### *Book Chapters:*

1. C. Liu, **X. Sheng**, “Transfer-Printed Devices for Biomedical Applications”, in *Transfer Printing Technologies and Applications* ed. by C. Cao and Y. Sun, Elsevier (2024).
2. H. Ding, **X. Sheng**, “Thin-Film III-V Single Junction and Multijunction Solar Cells and Their Integration onto Heterogeneous Substrates”, in *Inorganic Flexible Optoelectronics: Materials and Applications* ed. by Z. Ma and D. Liu, Wiley-VCH (2019).
3. **X. Sheng**, S. Wang, L. Yin, “Flexible, Stretchable and Biodegradable Thin-Film Silicon Photovoltaics”, in *Advances in Silicon Solar Cells* ed. by S. J. Ikhmayies, Springer-Verlag (2018).
4. L. Yin, **X. Sheng**, “Nonconventional Biosensors Based on Nanomembrane Materials”, in *Nanobiomaterials: Classification, Fabrication and Biomedical Applications* ed. by X. Wang, M. Ramalingam, X. Kong and L. Zhao, Wiley-VCH (2018).
5. **X. Sheng**, *Thin-film Silicon Solar Cells: Photonic Design, Process and Fundamentals*, LAMBERT Academic Publishing (2012).

### *Patents:*

1. 盛兴, 王华春, 基于光电调控的三维生物支架, ZL 2023 1 0140332.9
2. 盛兴, 孔德颖, 尹斓, 一种钙钛矿单晶薄膜的制备方法及其应用, ZL 2022 1 0657214.0
3. 盛兴, 赵钰, 谢杨, 植入式探针及其制备方法、控制电路及控制系统, ZL2021 1 0376512.8
4. 盛兴, 赵钰, 谢杨, 植入式探针、控制电路及控制系统, ZL2021 2 0717664.5
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10. **X. Sheng**, J. Liu, J. Michel, A. M. Agarwal, L. C. Kimerling, “Pseudo-periodic structure for use in thin film solar cells”, US20100307579 / WO2010141145.

### Services

#### *Internal at Tsinghua:*

- Panelist in postdoc searching committee
- Panelist in graduate admission committee
- Panelist in undergraduate admission committee
- Panelist in graduate thesis committee
- Freshmen Mentor
- Supervising undergraduate students supported by the Student Research Training (SRT) program

#### *External:*

- *Society Membership*
  - *Optica Fellow*
  - *IEEE senior member*
  - *SPIE life member*
- *Journal Editor*
  - *Optical Materials Express*, Associate Editor, 2017–2023.
  - *Fundamental Research*, Special issue “Multimodal Technologies for Neural Modulation and Sensing”, Lead Editor, 2023.
  - *IEEE Journal of Selected Topics in Quantum Electronics*, Special issue “Flexible Optoelectronics”, Guest Editor, 2023.
  - *Frontiers in Nanotechnology*, Feature issue “New Technologies for Large-Scale Recording and Modulation in the Brain”, Guest Editor, 2021.
  - *Optical Materials Express*, Feature issue “Bio-inspired and Bio-integrated Photonic Materials and Devices”, Lead Editor, 2019.
- *Board Member*
  - *Chinese Association of Automation*
  - *Chinese NeuroScience Society*
  - *Chinese Society of Biomedical Engineering*
- *Conference Organizer for multiple domestic and international conferences*
  - 2024 Nature Conference on Transformative Technologies for Neuroengineering, Shenzhen, China. Co-Organizer.
  - 2023 Optica Advanced Photonics Congress, Solar Energy and Light-Emitting Devices (SOLET) Topical Meeting, Busan, Korea. Subcommittee.
  - 2020 CIMTEC 9th Forum on New Materials, Montecatini Terme, Italy. International Advisory Board Member.
  - 2019 IEEE-EMBS 16th International Conference on Wearable and Implantable Body Sensor Networks (BSN), Chicago, IL, USA. Technical Program Committee.
  - 2019 MRS spring meeting, Phoenix, AZ, USA. Symposium Organizer.
  - 2017 OSA IPR meeting, New Orleans, LA, USA. Subcommittee.
  - 2016 MRS fall meeting, Boston, MA, USA. Symposium Organizer.
  - 2016 MRS spring meeting, Phoenix, AZ, USA. Symposium Organizer.
- *Reviewer for multiple international journals*
- *Proposal Reviewer for NSFC, and multiple international funding agencies*
- *Co-president, MIT Chinese Association of Science and Technology, 2010.*
- *Scientific consultant for several high-tech start-up companies*

## **Awards and Honors**

### *International*

- Optica Fellow, 2024
- Rising Stars of Light (Finalists), by Light: Science & Applications, 2023

- Young Scientist Award, Photonics & Electromagnetics Research Symposium (PIERS), 2018
- Young Scientist Award, Microsystems & Nanoengineering Summit (MINE), 2018
- Best Poster Award (2nd prize) in Nature Conference on Flexible Electronics, Nanjing, 2016
- Gordon Engineering Leadership Teaching Assistantship, MIT, 2011
- Energy Initiative Seed Fund Award, MIT, 2010
- Best Poster Award (runner-up) in the 35th IEEE Photovoltaic Specialists Conference, 2010
- Energy Initiative Martin Fellowship, MIT, 2010
- DuPont-MIT Alliance Fellowship, 2007
- MIT Presidential Fellowship, 2007

*Domestic (in Chinese)*

- 2024, 国家自然科学基金委交叉科学部杰出青年科学基金
- 2024, 《中国激光》“生物医学光子学-最美封面论文”
- 2022, 清华大学, 电子工程系, 郑君里教书育人优秀教师奖
- 2022, 清华大学, 电子工程系, 周炳琨学者奖
- 2022, 中国生物医学工程大会, 青年优秀论文报告
- 2021, 中国材料研究学会, 科学技术一等奖(基础研究类)(排名第三)
- 2019, 《中国激光》主编推荐奖优秀论文
- 2018, “中国新锐科技卓越影响奖”
- 2014, 国家海外人才计划青年项目
- 2007, 清华大学, 优良毕业生
- 2006, 清华大学, 杜邦学生奖学金
- 2005, 清华大学, 三星学生奖学金
- 2004, 清华大学, 伍占德学生奖学金
- 2003, 清华大学, 新生奖学金